Mining for the future: Addressing liabilities and unlocking sustainable transition opportunities

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In Summary

A twin transition is driving a surge in mineral demand as decarbonization and digitalization are reshaping the industrial base simultaneously. Energy systems are shifting toward electrification, renewables, storage and expanded grids, while artificial intelligence is driving heavy investments in data centers and computing infrastructure. Demand for critical minerals is set to rise accordingly. The International Energy Agency (IEA) projects that by 2040 lithium demand could increase fivefold; graphite and nickel may double; cobalt and rare earth elements could rise by 50–60% and copper demand by around 30%. Meeting this surge will require not only new mines, but expanded processing capacity, infrastructure and skilled labor. Meanwhile data-center electricity demand alone is expected to more than double by 2030, reinforcing the case for grid investment and additional materials.

Supply is struggling to respond. Although a more circular economy may eventually ease reliance on primary extraction, supply chains for key transition minerals remain slow to scale. New projects take years to permit and finance, and face rising environmental and social standards. The pace at which mining and refining capacity can grow is therefore a central constraint; failure to scale responsibly risks prolonging fossil-fuel dependence.

Environmental pressures reinforce these constraints. Mining contributes around 2–4% of global GDP, supports millions of livelihoods and accounts for approximately 4–7% of global greenhouse gas emissions. Although less significant than agriculture or urban expansion in driving forest loss, mining directly deforested nearly 20,000 km^2 between 2001 and 2023, generating roughly 0.75 Pg of CO_2 – slightly more than Germany's annual emissions in 2024. Stronger operational practices, rehabilitation and credible compensation mechanisms such as biodiversity offsets and reforestation could reduce these impacts. While debated, effective implementation could help limit net forest loss as mineral demand rises.

Bridging this gap will require substantial capital. We estimate cumulative investment needs of around USD1.1trn to 2040 across mining, processing and circularity. Sustainability has become a binding production constraint: permitting, insurance and community consent now shape capacity as directly as technical factors. Company disclosures suggest roughly USD450bn may be required for decarbonization, tailings manage-

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munities and customers would allow costs to be priced in upfront, streamline engagement and improve the credibility of approvals.

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